

E-Commerce Conversion Prediction Challenge

Objective

Predict whether a user converts using demographic, browsing behavior, traffic source, device, and purchase history features.

Data Analysis

The dataset contained 10,000 training samples. Missing values were present in Age, Income, and Time_on_site . The target variable was imbalanced, with more non-converted than converted users. Exploratory analysis indicated that discount exposure and referral traffic were associated with higher conversion rates.

Preprocessing

(Median imputation for missing values , Standard scaling of numerical features , One-hot encoding of categorical variables , Stratified train-validation split)

F1 Score used for evaluation

Models Evaluated

Model	F1 Score
Logistic Regression	0.381
Random Forest	0.384
Balanced Random Forest	0.511
Tuned Balanced Random Forest	0.566

Final Model

- Tuned Balanced Random Forest: (n_estimators = 500 , max_depth = 10 , min_samples_split = 5 ,class_weight = balanced)

Results

Validation F1 Score: 0.5659

Public Test F1 Score: 0.5266

Conclusion

The Tuned Balanced Random Forest achieved the best performance and was used to generate predictions for the private test set. The model effectively handled class imbalance and improved conversion prediction accuracy compared to baseline approaches.